

WASP-36b

R-0675

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-36_260211 · created 2026-07-27T18:58:16+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2461082.8202 +/- 0.0063 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.103 +/- 0.048
Transit depth (Rp/Rs)^2	1.05 +/- 0.99 [%]
Orbital Inclination (inc)	87.64 +/- 2.89
Airmass coefficient 1 (a1)	0.34 +/- 0.3
Airmass coefficient 2 (a2)	0.79 +/- 0.66
Scatter in the residuals of the lightcurve fit is	101.85 %
Best Comparison Star	#3 - [424, 133]
Optimal Method	PSF photometry
Transit Duration (day)	0.086 +/- 0.011

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R* fit vs published	0.103 ± 0.048 vs 0.1368 (deviation 0.34σ)
Duration fit vs published	0.086 d vs 0.0758 d (deviation 0.45σ)
Mid-time O-C	-15.5 min
Depth SNR	1.0
Residual scatter	101.85 %

Light curve

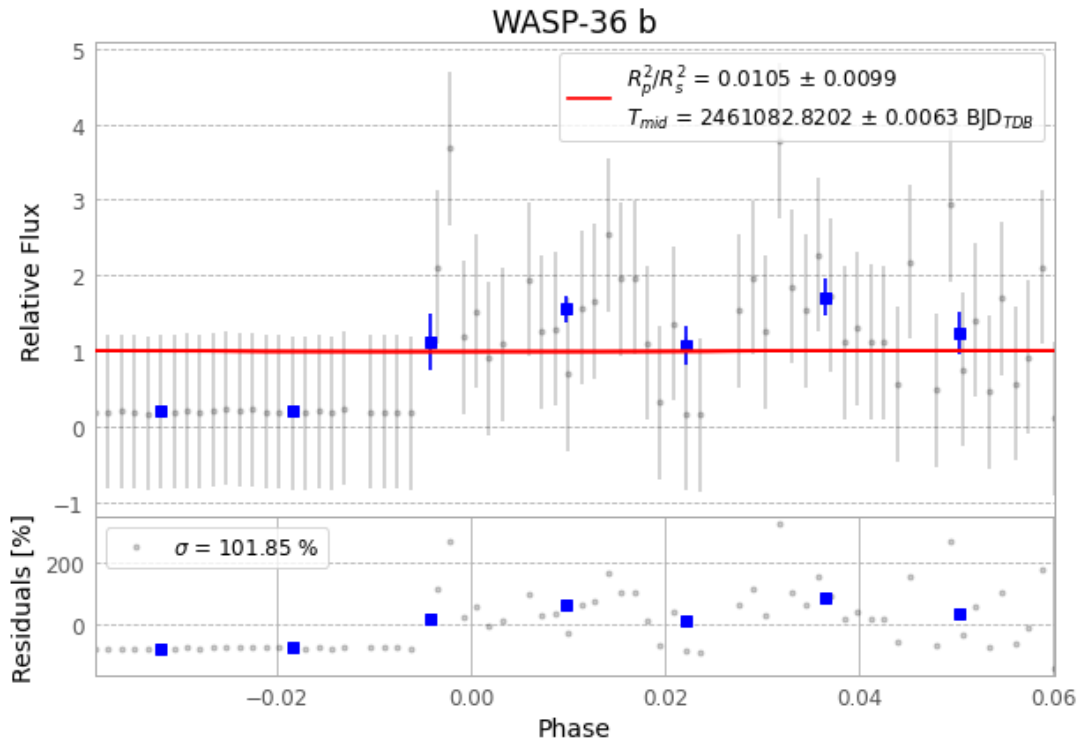


Figure 1 — FinalLightCurve_WASP-36 b_2026-02-10.png (SHA-256 2c016323f1cde44d...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [356, 165], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 a668ac56c96eb6ac...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 62f7813

Data provenance

74 FITS frames (49 MB), WASP-36260211060616.FITS → WASP-36260211094515.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-36-260211.log (41fe8045865a...), FinalLightCurve_WASP-36 b_2026-02-10.png (2c016323f1cd...), FinalLightCurve_WASP-36 b_2026-02-10.csv (2b42014d1d2c...)

Compute resources

Wall time 200.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-27 18:58Z.